

# Agentic Onboarding — Risk & Fraud Capture at STP

One Onboarding Risk Assessment Agent turns the first customer touch into the underwriting moment — reading the risk, catching the fraud, and deciding the next step, all before the policy is issued.

## Safe STP rate ▲

auto-issued on evidence, not just a clean form

## Manual underwriting ▼

risk & fraud captured up front, not re-worked later

## Claims leakage ▼

adverse selection & fraud caught pre-issue

## Fraud caught ▲

at onboarding — not at the claim

### The challenge

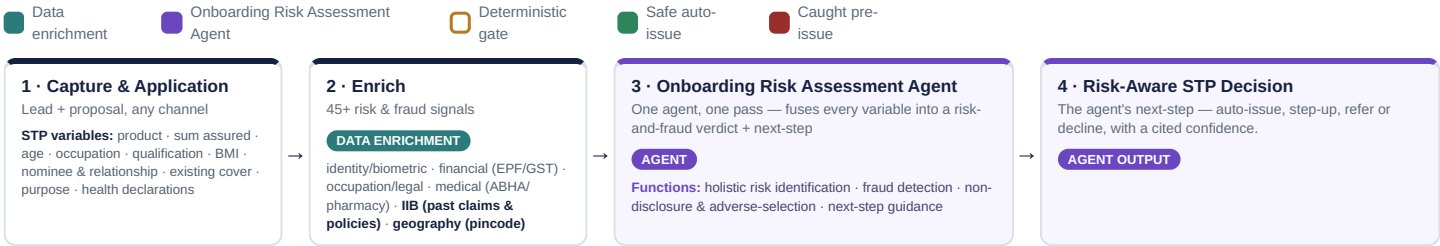
Insurance is increasingly won or lost at onboarding. The insurers pulling ahead read a proposal's true risk — and its fraud — at the very first touch, and act on it while the customer is still on the screen, with no extra manual underwriting. Do it blind, and the same risk resurfaces at the worst possible moment: as a large, avoidable claim.

### The solution

Aurum makes onboarding itself the underwriting moment. A single **Onboarding Risk Assessment Agent** fuses every application field and enrichment signal — identity, financial, medical, occupation, IIB cross-industry history and geography — into one risk-and-fraud verdict and **guides the next step** — auto-issue clean cases instantly, and flag adverse selection, non-disclosure and fraud before the policy is issued.

## Where the intervention happens — the STP onboarding stage

One agent, one pass: it fuses every variable into a risk-and-fraud verdict and decides the next step — auto-issue, step-up, refer or decline.



The agent produces the risk-and-fraud verdict and the next-step decision. The deterministic STP eligibility screen (age band, SA limit, KYC/AML) still runs; the agent enriches it, it does not replace it.

SAFE → AUTO-ISSUE

### Instant digital policy

Clean, evidence-backed cases issue straight through — now with a risk-and-fraud evidence chain, so more can be auto-issued with confidence.

RISK / FRAUD CAUGHT — PRE-ISSUE

### Step-up · Refer · Decline

Adverse selection, non-disclosure or fraud detected → a verification step-up or a referral, with cited evidence. The bad risk never becomes a claim.

### THE SIGNAL SURFACE THE AGENT REASONS OVER

**Application variables:** age, occupation, product, sum assured, qualification level, BMI, nominee & relationship, purpose / source of funds.

**Occupation & legal:** hazard class, director defaults/MCA, litigation/ FIR, background verification.

**Geography:** pincode-level mortality, fraud-hotspot & catastrophe risk from the applicant's address.

**Identity & fraud:** PAN/Aadhaar, face-match + liveness, deepfake/ multi-face, mobile & email vintage + fraud, mobile-to-PAN/bank linkage, device/behaviour.

**Medical (low-friction):** ABHA records, pharmacy history, facial-scan vitals, BMI, declaration cross-check.

**Behavioural / adverse-selection:** cross-insurer application velocity, sudden cover-stacking, buying right after a health event.

**Financial:** EPF/GST income, banking & account vintage, source-of-funds, PEP/AML/sanctions.

**Cross-industry — IIB:** prior claims (incl. repudiations), existing policies & aggregate sum assured across all insurers.

**Fused output:** one risk-and-fraud verdict → the next-step decision (auto-issue / step-up / refer / decline), with a cited evidence chain.

# One Onboarding Risk Assessment Agent, four functions

A single agent captures the risk and fraud at onboarding and decides the next step — no separate fraud or non-disclosure agents to wire together.

## Onboarding Risk Assessment Agent

The one reasoning component on the STP path. It reads every application field and enrichment signal, forms a single risk-and-fraud verdict with a confidence score, and tells the workflow what to do next. The deterministic controls it depends on — STP eligibility gate, regulatory limits, AML hard-stops — stay deterministic; the agent supplies the judgment they can't.

| Function                               | What it does                                                                                                                                                                                                       |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 • Holistic risk identification       | Fuses identity, demographic, occupation, lifestyle, medical and financial signals — plus age, product, sum assured, qualification, BMI, geography and IIB history — into one risk view.                            |
| 2 • Fraud detection                    | Identity & synthetic-identity fraud (deepfake, liveness, multi-face, mobile-to-PAN mismatch), mule-account and new-vintage proxies, application misrepresentation, device/behavioural anomalies, PEP/AML.          |
| 3 • Non-disclosure & adverse-selection | Cross-checks declared vs enriched — undisclosed illness (ABHA/pharmacy), income vs sum assured, and IIB prior claims, existing policies & cross-insurer cover-stacking. Two agreeing signals raise the confidence. |
| 4 • Next-step guidance                 | Decides and drives the workflow: auto-issue, assign a loading class, step-up verification, refer to a human, or decline — each with a cited reason.                                                                |

The agent classifies the risk and decides the next step; the STP eligibility gate and AML hard-stops stay deterministic.

## REAL-WORLD USE CASE 1 — ADVERSE SELECTION + FRAUD

# The case a deterministic STP rule would have auto-issued

A low sum-assured application that looks clean on the form — and would have become an early claim.

**Applicant — Suresh Kumar, 38.** Applies on WhatsApp for a **₹25 lakh term plan**, 20-year term (age 38 · product term · SA ₹25 L). Low SA → **STP-eligible under a deterministic low-SA rule** (no medical). Declares: **healthy, non-smoker, salaried graduate, no existing cover**. On the form, nothing to see.

| Onboarding step    | Variables / Inputs (IN)                                                                                        | Point               | Outputs (OUT)                                                                                                                                                                                                                                                         |
|--------------------|----------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Capture            | product, SA ₹25 L, age 38, occupation (salaried), qualification (graduate), "no existing cover"; Mobile, Email | ENRICH              | Mobile recently ported (<30 days), email newly created → fraud proxy elevated                                                                                                                                                                                         |
| Identity + geo     | PAN, Mobile; address → pincode                                                                                 | ENRICH              | PAN valid but mobile-to-PAN link absent; address maps to a known high-fraud pincode                                                                                                                                                                                   |
| Employment         | declared salaried graduate                                                                                     | ENRICH              | Declared salaried, but no EPF record; income unverifiable → misrepresentation                                                                                                                                                                                         |
| IIB cross-industry | PAN / identity                                                                                                 | ENRICH · IIB        | IIB shows 3 other term policies across insurers in 45 days (aggregate ≈ ₹1.1 cr) + a prior repudiated claim                                                                                                                                                           |
| Risk assessment    | All variables fused                                                                                            | ORA AGENT           | Fraud + IIB + geography cluster → agent pulls a consented ABHA/pharmacy check despite low SA → oncology diagnosis + chemotherapy 2 months ago (undisclosed). Undisclosed illness + IIB cover-stacking + prior repudiation → high-confidence adverse selection + fraud |
| Next step          | Risk verdict                                                                                                   | ORA AGENT<br>CAUGHT | A low-SA rule would auto-issue; the agent classifies it adverse selection + fraud → blocks issuance → refer / decline with cited evidence; fraud pattern captured for the KB                                                                                          |

The cover-stacking and prior repudiation come straight from IIB — the strongest anti-selection signal, invisible on the applicant's own form. Clean low-SA cases still auto-issue instantly.

Illustrative worked example · figures indicative.

## Two more ways the same agent protects the book

Different variables, different risk types — mortality under-rating and financial crime — each invisible on the application form.

### Use case 2 — Priya Nair, 34 · hidden hazardous occupation + high BMI + undisclosed tobacco

Age 34 · product term · SA ₹40 L · declares "office administrator", non-smoker, healthy. Moderate SA → STP-eligible; would auto-issue at standard rates.

#### A DETERMINISTIC STP RULE WOULD

Auto-issue at **standard rates** — age, product and SA are in band; declared occupation is low-risk; no medical for this SA. Nothing trips a threshold.

#### THE AGENT CATCHES (VARIABLES)

**Occupation:** GST/MCA show she is a working partner in a fireworks unit (hazardous class). **BMI 31** from the application's own height/weight. **Tobacco:** pharmacy shows a repeat **smoking-cessation (nicotine-replacement) prescription** — inconsistent with "non-smoker". IIB is clean, but three material risk variables the form understated.

#### OUTCOME & VALUE

Not fraud — but **risk variables the form hid**. Assessed as standard, this is a materially **under-rated** risk vs a hazardous-class + high-BMI + smoker profile. The agent **classifies it into the correct loading class** (or steps it up) → caught at onboarding, not absorbed as mortality-cost leakage.

### Use case 3 — "Vikram R", 29 · money-mule / premium-laundering (AML)

Age 29 · product single-premium endowment · SA ₹30 L · ₹30 L paid upfront. Identity clean, SA in band → STP-eligible.

#### A DETERMINISTIC STP RULE WOULD

Auto-issue — **clean KYC, SA within limit, premium received**. A rule keyed on identity + product + SA + payment sees a completed sale.

#### THE AGENT CATCHES (VARIABLES)

**Qualification & income vs SA:** ₹30 L single premium on ₹6 L declared income and a modest qualification; premium from a **<45-day-old account**. IIB shows a burst of single-premium policies across insurers. Mobile-to-PAN/bank mismatch; PEP-adjacent.

#### OUTCOME & VALUE

A different "bad risk realised late" — an **AML / PMLA exposure and a laundering channel** (buy-then-early-surrender), not a mortality claim. The agent **routes to compliance / enhanced due diligence pre-issue** with cited evidence, instead of auto-issuing a laundering vehicle.

### Why a deterministic STP gate can't catch any of these three

On the few fields an STP gate reads — age band, sum assured, product, clean declarations, KYC/AML pass — **all three cases are immaculate**. The risk and the fraud live in the **relationships between variables no single field captures**: an undisclosed illness that only means something *alongside* IIB cover-stacking; a hazardous occupation that surfaces only in GST/MCA, not the declared job title; a premium whose size only looks wrong *against* verified income, qualification and account vintage. A rule can test a field; it cannot reason across variables it was never told to connect. The Onboarding Risk Assessment Agent does, and only escalates when the evidence clusters, so the clean majority still auto-issues instantly and the STP rate goes **up**, not down.

# The variables that drive the STP decision — and where each comes from

Every variable is captured from the application or an enrichment source, then fused by the Onboarding Risk Assessment Agent into the risk verdict and the next-step decision.

| Variable                              | Captured from                            | Enrichment / source                                | What it drives (agent function)                                                                                                 |
|---------------------------------------|------------------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Age                                   | Identity (PAN DOB); application          | PAN Profile · CKYC                                 | STP age-band gate; mortality baseline → risk profiling                                                                          |
| Occupation                            | Application; verified N4/N5              | Non-hazardous check · GST · MCA/Director           | Hazard class; moral hazard → risk · fraud                                                                                       |
| Product & plan                        | Application                              | —                                                  | Selects the STP ruleset & medical requirements → next-step                                                                      |
| Sum assured                           | Application                              | —                                                  | STP SA-limit gate; non-medical-limit trigger → next-step                                                                        |
| Qualification level                   | Application; corroborated                | Education / KYC profile                            | Non-medical limits, income credibility, moral hazard → risk · fraud                                                             |
| IIB — past claims & existing policies | Enrichment (identity/PAN)                | IIB (Insurance Information Bureau) query           | Prior claims/repudiations, existing policies, aggregate cover → cover-stacking, adverse selection, prior fraud → non-disclosure |
| Geography risk                        | Identity address → pincode               | Geo-risk indices (mortality · fraud-hotspot · CAT) | Location-risk weighting → risk · fraud                                                                                          |
| BMI                                   | Application (height/weight); facial-scan | Facial-scan vitals                                 | Medical risk; STP health eligibility & loading → risk                                                                           |
| Existing cover (declared)             | Application; cross-checked vs IIB        | IIB                                                | Insurable interest, over-insurance, non-disclosure → non-disclosure                                                             |
| Income                                | Application; verified                    | EPF / GST / banking                                | Affordability; SA justification (income multiple) → risk                                                                        |
| Nominee & relationship                | Application                              | —                                                  | Insurable interest; moral hazard → risk · fraud                                                                                 |
| Purpose / source of funds             | Application                              | AML · source-of-funds                              | AML / PMLA; moral hazard → fraud · next-step                                                                                    |
| Tobacco / lifestyle                   | Application; cross-checked               | Pharmacy (cessation scripts) · declarations        | Mortality loading; non-disclosure → risk                                                                                        |
| Family medical history                | Application                              | —                                                  | Hereditary risk on the mortality read → risk                                                                                    |

IIB = the Insurance Information Bureau of India's cross-industry repository — the one source that reveals what an applicant carries or has claimed with other insurers, which no single-insurer view or self-declaration can.  
Tobacco is inferred from pharmacy smoking-cessation / nicotine-replacement scripts and declaration cross-checks — not from a facial scan.

## Onboarding states — variables, enrichment & the agent's action

Each state lists the STP decisioning variables it contributes, the enrichment invoked, and what the single Onboarding Risk Assessment Agent does with it.

| State                             | STP decisioning variables captured                                                                                                                                                         | Data enrichment (source)                                                   | Onboarding Risk Assessment Agent                                                                    |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| N1 · Capture & Application        | product, sum assured, term, premium/mode, age, occupation, qualification, height/weight → BMI, nominee & relationship, existing cover, purpose / source of funds, tobacco & family history | Email/Mobile auth + fraud · Mobile age (silent)                            | Ingests all variables into the risk view                                                            |
| N2–N3 · Identity, KYC & Geography | age (verified DOB), gender, name, address → geography / pincode risk                                                                                                                       | PAN Profile · PAN–Aadhaar · CKYC · PIVC face + liveness · geo-risk indices | Fraud detection (identity/synthetic) + geography weighting                                          |
| N4 · Employment & Income          | verified occupation, employer, income, qualification (corroboration)                                                                                                                       | EPFO/PAN · EPF passbook · GST details                                      | Income vs sum assured, affordability & SA justification                                             |
| N5 · Occupation & Legal           | occupation hazard class, director defaults, PEP, litigation/FIR                                                                                                                            | Non-hazardous · MCA/Director · PEP/AML · litigation BI · FIR               | Moral-hazard & fraud signals into the verdict                                                       |
| N6 · Banking                      | premium-payment viability, account name & vintage, source-of-funds proxy                                                                                                                   | Reverse penny drop · mobile-to-bank · silent bank A/C                      | Mule-account & laundering proxies                                                                   |
| N7 · Medical (low-friction)       | BMI, vitals, ABHA diagnoses, pharmacy, tobacco corroboration                                                                                                                               | Facial-scan vitals · ABHA records · pharmacy history                       | Medical risk & non-disclosure cross-check                                                           |
| Cross-industry · IIB              | prior claims (incl. repudiated), existing policies, aggregate SA across insurers, application velocity                                                                                     | IIB (Insurance Information Bureau) query                                   | Adverse-selection & cover-stacking, prior-fraud detection                                           |
| Risk-Aware STP Decision           | fused: age · occupation · product · SA · qualification · geography · BMI · income · existing cover · IIB history                                                                           | Reuses all facts (no new pulls)                                            | Agent — next-step: auto-issue / loading class / step-up / refer / decline, with a cited confidence. |

**Reading the spec.** A deterministic STP gate uses a handful of these variables as thresholds (age band, SA limit, product). Aurum keeps those deterministic — the single Onboarding Risk Assessment Agent adds the reasoning across the rest (IIB history, geography, occupation-vs-declared, income-vs-SA, BMI, source-of-funds) and guides the next step. Camera-dependent states (PIVC, facial-scan) need App/Web or WhatsApp media; OTP/consent states work on every channel.

# Claims-leakage avoided & loss-ratio impact

Onboarding-stage assessment converts claims-stage losses into pre-issue decisions. Illustrative, for a mid-size private life insurer; every driver is an editable assumption.

≈ ₹13.4 cr

gross claims-leakage avoided / yr

≈ ₹11.6 cr

net benefit / yr (after Aurum cost)

≈ 7.4×

return on onboarding-screening spend

≈ 110 bps

off the new-business loss ratio

| Driver (blue = editable assumption)                                                                                 | Value                           |
|---------------------------------------------------------------------------------------------------------------------|---------------------------------|
| New policies issued / year                                                                                          | 4,80,000                        |
| Average new-business sum assured                                                                                    | ₹35 lakh                        |
| Bad-risk auto-issues prevented — fraud + material non-disclosure + adverse selection a deterministic STP would pass | 0.10% → 480 cases               |
| Avoidable-claim conversion within the contestability window                                                         | 8% → 38 claims                  |
| Average payout avoided per case (repudiable fraud / non-disclosure → full SA)                                       | ₹35 lakh                        |
| Gross claims-leakage avoided                                                                                        | ≈ ₹13.4 cr / yr                 |
| Less: Aurum onboarding assessment (₹38 / case × 4,80,000)                                                           | (₹1.8 cr)                       |
| Net benefit                                                                                                         | ≈ ₹11.6 cr / yr · ≈ 7.4× return |
| New-business loss-ratio impact (₹13.4 cr on ≈ ₹1,200 cr first-year premium)                                         | ≈ 110 bps improvement           |

*Illustrative and deliberately conservative: it counts only the **fraud / non-disclosure / adverse-selection** slice a deterministic STP would have auto-issued — not the wider manual-underwriting savings, cycle-time or conversion gains, nor softer avoided costs (claims investigation & litigation, AML / regulatory penalties, reputational damage). Scale linearly with book size; every driver is a lever the insurer can set from its own experience.*

## Where the value sits — the asymmetry

| Onboarding outcome                                    | Share of proposals | What it saves                   |
|-------------------------------------------------------|--------------------|---------------------------------|
| Clean → safe auto-issue                               | ~60–65%            | manual UW touch avoided         |
| Risk / fraud clustered → step-up or refer, pre-issue  | ~5–8%              | the large claims losses avoided |
| Remaining → routed to detailed underwriting as before | ~30%               | unchanged                       |

### More safe STP

Auto-issue on an evidence-backed verdict — more clean cases straight through, with confidence.

### Less manual UW

Only clustered-risk cases are stepped up — no blanket manual review.

### Lower claims leakage

Adverse selection, non-disclosure & fraud caught pre-issue.

### Auditable by design

Every decision, block or issue ships with a cited evidence chain for IRDAI / PMLA.